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Institutional conflicts of interest in regional water management: a case study of the water supply in the Taipei metropolitan area

Chieh-Ying Chen¹ , Yu-Syuan Cai¹, Yu-Ju Hung^{2*} and Gene Jiing-Yun You^{1*}

Abstract

Institutional conflicts of interest constrain regional water management in metropolitan systems supplied by multiple reservoirs and operated by separate utilities. Such conflicts arise when decentralized utilities pursue cost-minimizing strategies that diverge from system-wide reliability and social welfare objectives. This study examines these dynamics in the Taipei metropolitan water supply system, where two reservoirs with contrasting hydrological reliability are physically connected through a cross-basin water transfer infrastructure intended to rebalance regional allocation. A deterministic scenario-based analysis using the Water Evaluation and Planning (WEAP) model is conducted under varying hydrological stress regimes to evaluate alternative allocation priorities and pricing structures. The analysis links hydrological conditions, infrastructure operation, and utility-level financial incentives to system-level social loss. Results indicate that pricing and transfer-payment adjustments can align institutional incentives with socially preferred outcomes when scarcity is localized and overall system reliability is maintained. However, when severe drought affects the historically more reliable subsystem, pricing mechanisms alone become insufficient to prevent socially suboptimal allocation. These findings demonstrate that physical integration without incentive alignment does not guarantee effective coordination and clarify the conditions under which pricing instruments can mitigate institutional conflict in regional water systems.

Keywords Water pricing, Institutional conflicts, Inter-basin water transfer, Financial interest, Self-interested behavior

Introduction

Climate change and increased human activity have led to conflicts over water resources at local, regional, and global scales (Anderson et al. 1997; Ayling and Kelly 1997; Buckles 1999). Human interventions such as land-use change, reservoir regulation, irrigation expansion, and groundwater abstraction have substantially altered natural hydrological processes, thereby amplifying water scarcity under a changing climate (Pokhrel et al. 2017). In parallel, climate change has increased the frequency and intensity of extreme weather events, including droughts and heatwaves, further exacerbating water shortages and disrupting existing water allocation systems (Clarke et al. 2022). Lynch (2012) reported that climate change

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can increase water demand and intensify the need for its reallocation. As a result, water–society interactions are evolving rapidly as climate-induced scarcity manifests through heightened competition for water at local, national, and international levels. Despite the interconnected nature of hydrological processes within a watershed (Johnson et al. 1981), water management practices remain fragmented across sectors and institutions, limiting the capacity to respond effectively to escalating climate-related water stress (Mitchell 2005).

Integrated Water Resource Management (IWRM) was developed to address conflicts over water resources by promoting coordinated decision-making across sectors and administrative levels (Agarwal et al. 2000; Biswas 2004; Thomas and Durham 2003; Mitchell 2005). IWRM seeks to bring together multiple stakeholders to overcome inefficiencies that arise when water resources are managed in isolation, and is commonly implemented through watershed authorities or cooperative arrangements among existing agencies. However, while IWRM emphasizes coordination, it does not inherently provide institutional mechanisms to resolve collective-action problems or reconcile conflicting interests, particularly when decisions involve asymmetric power relations or the protection of minority stakeholders.

Recent governance research highlights that the effectiveness of coordinated water policies depends not only on institutional integration but also on how governance systems respond to non-linear interactions among actors, rules, and policy instruments (Bolognesi and Fischer 2025). Weak horizontal and vertical coordination can therefore undermine policy outcomes even in formally integrated frameworks. Dinar and colleagues (Dinar et al. 1997; Dinar 2000, 2007) provided extensive analyses of water conflict and cooperation, emphasizing the roles of treaties, negotiations, and institutional design. More recently, Dinar (2024) argued that effective water resource management requires closer alignment between economic instruments and modeling tools to support coordination across institutions and scales.

Governments play a central regulatory role in IWRM, yet their effectiveness is strongly influenced by the distribution of political power among stakeholders (Fisher and Huber-Lee 2012; Dinar et al. 2007). While user-based allocation mechanisms often provide greater flexibility than centralized allocation, their success depends on institutional capacity for collective action. Market-based instruments, such as tradable water rights, may facilitate inter-sectoral reallocation by creating incentives for efficient water use, but their performance ultimately hinges on coordinated governance arrangements that integrate regulatory oversight, economic valuation, and system-level analysis (Dinar et al. 2007; Kurian and McCarney 2010; Fisher and Huber-Lee 2012).

Water-related institutions can provide incentives and determine outcomes in both individual and collective decisions related to water development, allocation, use, and management (Saleth and Dinar 2005; Koundouri and Papandreou 2013; Amarasinghe and Sharma, 2008). They play a pivotal role in resolving conflicts and promoting cooperation in water allocation and management (Gleick and Heberger 2012). Institutional conflicts associated with water resources are often more persistent and complex than purely technical problems. Laws and regulations governing the distribution of benefits and costs establish the ground rules for conflict resolution, and institutional factors can strongly influence the outcomes of water resource programs. Unanticipated institutional conflicts may therefore undermine program implementation (Di Gregorio et al. 2008; Breit et al. 2012).

The fundamental principle of water allocation is to maximize social welfare through the efficient use of water resources. However, water resources may be state-owned, privately owned, or treated as common property. Many researchers have focused on property rights to surface water (Burness and Quirk 1979; Johnson et al. 1981; Colby et al. 1993). Burness and Quirk (1979) argued that inefficiencies can arise under the doctrine of prior appropriation due to unequal risk-sharing among water users, which can hinder innovation and result in inefficient water use. Water rights specify how water is divided among and within sectors (Holden and Thobani 1996; Johansson 2000; Johansson et al. 2002).

From the perspective of economic efficiency, market-based mechanisms are often recommended as instruments for water allocation and conflict mitigation (Dinar et al. 1997; Rosegrant and Binswanger 1994). Water pricing allows scarcity signals to guide water allocation decisions, with higher prices reducing consumption and lower prices encouraging increased use. Consequently, maintaining artificially low water prices through subsidies may intensify pressure on water resources. Given diminishing marginal returns to water use, pricing has been widely recognized as a policy instrument that can promote both efficiency and conservation (Anderson 1983; Anderson and Hill 1997; Lowry 2003).

Recent studies extend this perspective by linking pricing strategies with conflict resolution mechanisms. Leader–follower and game-theoretic frameworks demonstrate that pricing policies, when combined with structured negotiation and stakeholder coordination, can reduce conflicts and improve allocation outcomes in water-scarce basins (Jahanshahi et al. 2023; Kamyab et al. 2024). Moreover, market-based and real-time pricing approaches that account for hydrological uncertainty can enhance transparency and adaptability in basin-wide water allocation, thereby mitigating disputes arising from asymmetric information and uncertain supply conditions

(Vahedizade et al. 2023). These studies highlight that water crises are not solely driven by physical scarcity but are also shaped by institutional and legal arrangements characterized by unclear property rights and centralized decision-making (Anderson and Hill 1997; Franks et al. 2014).

Transboundary water conflict arises when water resources are allocated along administratively defined boundaries that do not align with hydrological systems, leading to unequal access and competing claims (Zeitoun and Mirumachi 2008). Because many watersheds extend across multiple jurisdictions, the costs and benefits of water management decisions are unevenly distributed, particularly between upstream and downstream users (Ananda 2013). Governance approaches that emphasize stakeholder interests and negotiated allocation mechanisms have therefore become increasingly important.

In urban contexts, water governance challenges are further intensified by dense infrastructure networks and multi-actor interdependencies. Governance theory highlights the need for integrated, adaptive, and participatory frameworks that operate across administrative and sectoral boundaries to enhance urban water resilience (Saikia et al. 2022). From an eco-water security perspective, sustainable water management requires governance arrangements that balance ecological integrity with socio-economic objectives through cross-scale coordination (Xia et al. 2021).

This study examines institutional conflicts in the regional water supply system of northern Taiwan. The region is supplied by two major reservoirs, Feitsui Reservoir and Shihmen Reservoir, which serve Taipei City, New Taipei City, and Taoyuan City. Water supply in the region is managed by two separate utilities operating

under different institutional and financial arrangements. Because the hydrological regimes and supply responsibilities of the two reservoirs differ substantially, the regional water allocation structure exhibits significant asymmetry. To mitigate this imbalance, an inter-system water transfer project, the BanHsin Water Supply Project, was constructed to enable water transfer from the Feitsui system to areas traditionally supplied by the Shihmen system.

Despite the availability of transfer infrastructure, the operation of the BanHsin Water Supply Project has remained limited in practice. Independent cost structures and revenue mechanisms may discourage water transfers even when such transfers would improve regional water allocation efficiency. This situation illustrates how institutional incentives can constrain the effective use of existing water infrastructure.

This study, therefore, examines how pricing mechanisms and financial incentives influence water allocation decisions between the two water utilities. The Water Evaluation and Planning System (WEAP) is employed to simulate alternative allocation schemes under different hydrological conditions. By comparing the interests of water utilities with broader social welfare outcomes, this study evaluates whether pricing adjustments can reduce institutional conflicts and improve regional water allocation. The following section describes the study area and the modeling framework used in the analysis.

Methodology and materials

Water allocation conflicts in the study area

This study focuses on the regional water supply system of northern Taiwan, as shown in Fig. 1, which supports urban water demand in Taipei City, New Taipei City, and Taoyuan City, as well as agricultural irrigation demand

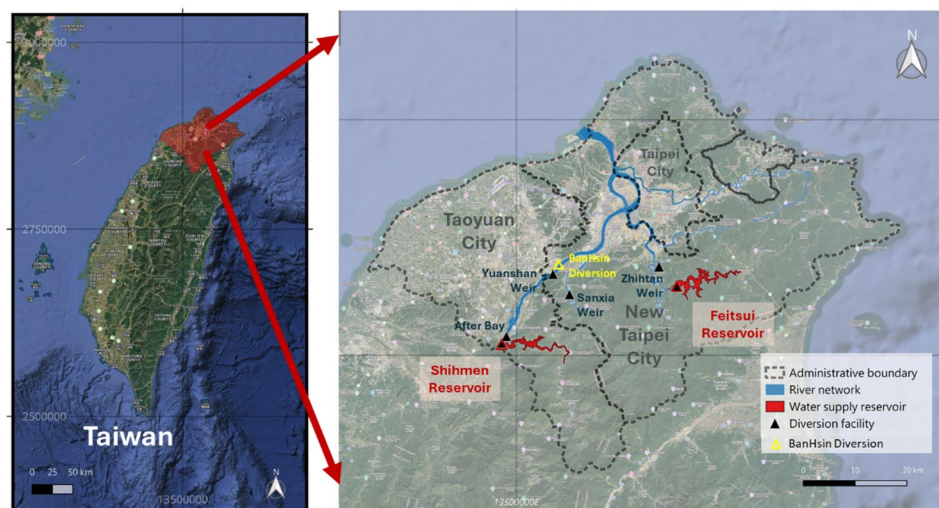


Fig. 1 Study area and major water supply infrastructure in northern Taiwan. The map shows administrative boundaries, river networks, water supply reservoirs, and diversion facilities, including the BanHsin Diversion connecting the Feitsui and Shihmen reservoir systems

primarily in Taoyuan. As Taiwan's political and economic center, this metropolitan region has a high population density and intensive urban development. Consequently, the stability of regional water allocation is essential.

As shown in Fig. 1, the regional water supply system is mainly supported by two major reservoirs, Feitsui Reservoir and Shihmen Reservoir. Feitsui Reservoir primarily supplies urban water to Taipei City. In this study, urban water supply refers to municipal and industrial (M&I) water use. In contrast, Shihmen Reservoir provides urban water to Taoyuan City and parts of New Taipei City and serves as a major source of irrigation water for agricultural areas in northern Taiwan.

Due to differences in watershed conditions and water demand patterns, the operational characteristics of the two reservoirs differ substantially. Feitsui Reservoir is located in a watershed influenced by the northeast monsoon, which provides relatively consistent rainfall even during the winter dry season. In addition, strict watershed protection regulations restrict land development within the catchment, helping maintain relatively reliable inflow conditions. As a result, the Feitsui system primarily serves the urban water supply of Taipei City. In contrast, the watershed of Shihmen Reservoir experiences greater rainfall variability (Wen et al. 2025). Shihmen Reservoir simultaneously supports urban water demand in Taoyuan and parts of New Taipei City, as well as

agricultural irrigation demand, resulting in significantly higher supply pressure. Furthermore, long-term sediment accumulation in Shihmen Reservoir has gradually reduced its effective storage capacity, thereby increasing the system's vulnerability to water shortages during drought periods.

From the perspective of reservoir operation, the ratio of annual water supply to effective storage capacity is typically below one for Feitsui Reservoir, whereas the corresponding ratio for Shihmen Reservoir is approximately four to five. This contrast indicates a clear imbalance in supply burden between the two reservoirs. In practice, Shihmen Reservoir must be accessed repeatedly each year to satisfy water demand in Taoyuan City and New Taipei City. In contrast, the Taipei supply system can first utilize natural flows from the Nanshih River upstream of Feitsui Reservoir before purchasing stored water released from Feitsui Reservoir. As illustrated in Fig. 2, these river flows are diverted through downstream diversion structures supplying Taipei's water treatment facilities. Consequently, stored water in Feitsui Reservoir is typically used only when river flows become insufficient, and the reservoir rarely needs to release large volumes of stored water to satisfy Taipei City's demand.

To mitigate this imbalance, the government implemented the BanHsin Water Supply Project, which enables water transfers from the Feitsui reservoir system to the

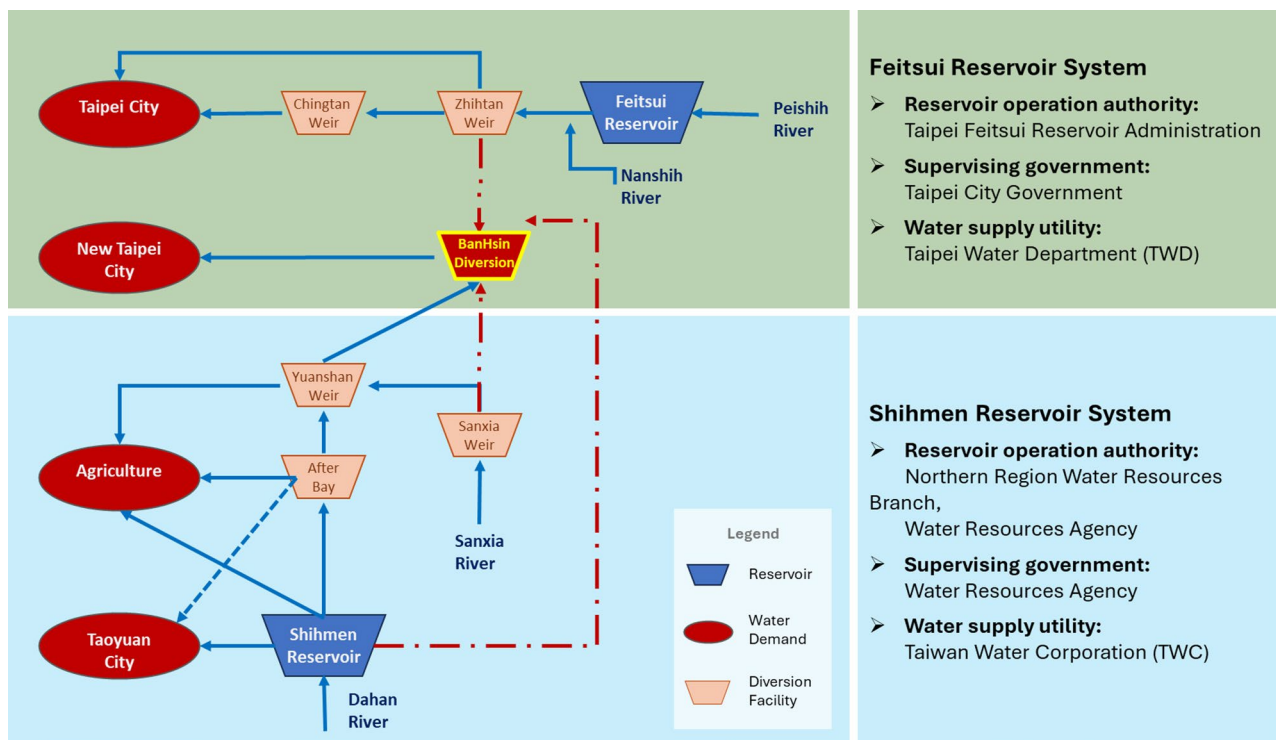


Fig. 2 Institutional structure of the Feitsui and Shihmen reservoir supply systems in northern Taiwan. The diagram illustrates major demand areas, diversion facilities, and institutional responsibilities. Green and blue backgrounds represent supply areas served by different water supply utilities

BanHsin service area in New Taipei City, a major municipal water demand area that has historically relied on the Shihmen reservoir system. The inter-system transfer is achieved through the BanHsin Diversion Plant, indicated by the yellow triangle in Fig. 1. This inter-system transfer infrastructure is intended to reduce supply pressure on Shihmen Reservoir and release additional capacity to support urban and agricultural demand.

However, regional water allocation is not determined solely by hydrological conditions and engineering infrastructure; institutional arrangements also play a critical role. In Taiwan, water rights are granted and regulated by the central government under the Water Act. The Water Resources Agency of the Ministry of Economic Affairs is responsible for reviewing water rights applications, determining withdrawal quantities, registering water rights, and managing overall water allocation to ensure that water resources are used in accordance with public interests and basin-level management objectives.

In practice, the institutions responsible for water rights governance are not the same as those responsible for operating urban water supply systems. As illustrated in Fig. 2, the Feitsui and Shihmen Reservoir systems belong to different administrative and operational frameworks. Feitsui Reservoir is operated by the Taipei Feitsui Reservoir Administration, with raw water supplied through the Taipei Water Department. In contrast, Shihmen Reservoir is operated by the Water Resources Agency, and its raw water is primarily delivered through the Taiwan Water Corporation supply system.

Under this institutional structure, the water supply utilities operate under independent financial and cost-accounting mechanisms. When each utility makes operational decisions based on cost-minimization considerations, inter-system water transfers may not necessarily occur even when they are hydrologically and technically feasible. In other words, the physical capacity for water transfer does not automatically translate into coordinated regional water allocation.

Therefore, the regional water supply system in northern Taiwan represents a case in which hydrological asymmetry between reservoirs interacts with fragmented institutional governance. This interaction creates potential

conflicts in regional water allocation, which form the central focus of this study.

Water evaluation and planning system (WEAP)

To examine these allocation dynamics and evaluate the effects of alternative pricing and operational strategies, this study employs the Water Evaluation and Planning (WEAP) system, developed by the Stockholm Environment Institute (SEI), as an integrated decision-support platform for water resources planning and policy analysis (Yates et al. 2005; Sieber and Purkey 2007; SEI, retrieved 2018).

WEAP has been widely applied to simulate basin-scale water balances and evaluate water management strategies across a broad range of regional contexts, including studies of climate change impacts on water availability in Mediterranean and semi-arid regions (Hadri et al. 2022; Moghadam et al. 2022), assessments of water pricing as a policy instrument for demand management (Jahanshahi et al. 2023; Mosavi et al. 2024), and coupled hydrological-behavioral modeling frameworks (Liu et al. 2022). These applications demonstrate that WEAP provides a flexible and effective framework for analyzing allocation strategies and supporting policy-relevant water management decisions under varying hydrological and institutional conditions.

The core of WEAP operates on the principle of mass conservation. The system represents a water supply network as a set of interconnected nodes and links, where nodes correspond to physical components such as reservoirs, river reaches, diversion structures, and demand sites, and links represent the conveyance infrastructure connecting them. At each time step, WEAP solves a linear programming (LP) problem to distribute available water among competing demand nodes. Allocation is governed by user-defined demand priorities and supply preferences, which together establish a hierarchical service order. High-priority demands, such as municipal water supply, are satisfied before lower-priority demands, such as agricultural irrigation. When multiple supply sources are available, supply preferences determine the sequence in which reservoirs, rivers, or diversion facilities are utilized. This priority-and-preference-based allocation logic allows the model to represent institutional rules and operational decision-making in a transparent and reproducible manner.

In this study, WEAP is employed as a deterministic, scenario-based allocation model to support comparative policy analysis rather than predictive simulation of future system states. The model is configured using the following categories of input data. Hydrological inputs include historical inflow time series for the major tributaries supplying both the Feitsui and Shihmen reservoir systems. Infrastructure inputs specify reservoir storage

Table 1 Annual water demand in the study area

Demand Site	Water Demand (million ton)
Agriculture Irrigation	344.32
Taipei City (M&I)	1074.67
Taoyuan City (M&I)	402.75
BanHsin service area, New Taipei City (M&I)	334.80

capacities, operating rules, diversion facility capacities, and conveyance links, including the BanHsin Diversion Plant. Demand inputs define annual M&I water demand for Taipei City, Taoyuan City, and the BanHsin service area in New Taipei City, as well as agricultural irrigation demand, as summarized in Table 1. Policy inputs encode the allocation priorities and supply preferences corresponding to each scheme examined in this study, as described in Sect. 2.3. The node-link structure of the WEAP model configuration for the study area is illustrated in Appendix Figure 8, which shows the six river reaches, seven reservoir and weir nodes, four demand sites, and one flow requirement node representing the minimum environmental flow at the Tamsui River outlet.

Water allocation schemes and hydrological scenarios

The allocation priorities and supply preferences introduced in Sect. 2.2 are evaluated across a range of hydrological conditions to assess system performance and institutional responses under varying levels of water stress. Climate change has been associated with increasing frequency and intensity of extreme drought events, heightening the risk of water shortages in water supply systems that already operate under high demand pressure (Clarke et al. 2022; Pokhrel et al. 2017). As described in Sect. 2.1, the Shihmen system exhibits substantially higher supply variability and demand burden than the Feitsui system, making it particularly vulnerable to drought-induced shortfalls. Under a changing climate, the capacity to redistribute supply burden across reservoir systems becomes increasingly critical. To evaluate whether the BanHsin Water Supply Project can serve this adaptive function, five hydrological scenarios are constructed spanning normal operations to severe drought in either subsystem, as summarized in Table 2.

Scenarios 4 and 5 do not represent climate projections or probabilistic drought forecasts. Instead, they are hypothetical stress-test scenarios designed to examine the sensitivity of the regional allocation system and institutional responses under progressively reduced inflow conditions in the historically more reliable Feitsui subsystem. The inflow reduction factors of 0.8 and 0.5 were selected to represent moderate and severe perturbations, respectively, allowing evaluation of system behavior across a broad range of hydrological stress levels rather than prediction of specific future drought conditions. Scenario 5 should therefore be interpreted as an exploratory boundary-condition case intended to illustrate how extreme degradation of the Feitsui subsystem could weaken the effectiveness of pricing-based coordination mechanisms, rather than as a prediction of future hydrological conditions.

Table 2 Summary of the five hydrological scenarios

Scenario	Hydrological Condition	Basis
1	Normal conditions	Historical average, 1980–2006
2	Moderate drought in the Shihmen system	Historical record, 1995–1996
3	Severe drought in the Shihmen system	Historical record, 2002–2003
4	Moderate drought in the Feitsui system	Feitsui and Xindian River inflows $\times$ 0.8
5	Severe drought in the Feitsui system	Feitsui and Xindian River inflows $\times$ 0.5

Table 3 Summary of the five water allocation schemes

Scheme	Primary Supply Source for BanHsin Service Area	Diversion Capacity (ton/day)
I	Shihmen system	—
II	Feitsui system	530,000
III	Feitsui system	1,005,000
IV	Shihmen system	530,000
V	Shihmen system	1,005,000

To evaluate system performance under these conditions, five water allocation schemes are formulated, varying along two dimensions: the operational capacity of the BanHsin Diversion Plant and the priority assigned to the Feitsui versus Shihmen supply sources for serving the BanHsin service area. These combinations are summarized in Table 3.

The BanHsin Water Supply Project was designed in two stages. Stage 1 provides a diversion capacity of 530,000 tons per day, intended to partially supplement Shihmen supply for the core BanHsin service area. Stage 2 increases the capacity to 1,005,000 tons per day, intended to fully transition the BanHsin service area to the Feitsui supply system.

Based on these two capacity levels and the priority assigned to each supply source, five allocation schemes are formulated. Scheme I represents the baseline condition without activation of the BanHsin Diversion Plant, in which the BanHsin service area relies entirely on the Shihmen system via existing local sources including Sanxia Weir. Schemes II and III assign supply priority to the Feitsui system for the BanHsin service area, at Stage 1 and Stage 2 capacities respectively, thereby actively redistributing supply burden from the Shihmen system to the Feitsui system. Schemes IV and V retain the same diversion capacities but assign priority to the Shihmen system, activating the Feitsui diversion only when the Shihmen supply is insufficient. Each of the five schemes is simulated under all five scenarios, yielding a total of 25 scheme–scenario combinations that form the basis of the analysis.

Social loss and financial costs

To evaluate the performance of alternative water allocation schemes from both societal and institutional perspectives, this study defines two distinct objective functions: social loss and financial cost. Social loss quantifies the aggregate welfare impact of water shortages on all demand sectors, whereas financial cost captures the economic burden borne by each water utility under a given allocation scheme. The scheme that minimizes social loss is treated as the socially preferred outcome, while the scheme that minimizes financial cost reflects each utility's self-interested choice.

Social loss

Social loss is defined as the sum of water shortfalls across all demand sectors, weighted by the unit economic loss associated with each sector, as expressed in Eq. (1):

$$SL = \sum (\Delta q_i \times L_i) \quad (1)$$

where SL is total social loss (million NTD), Δq_i is the volumetric water shortfall at demand site i (million tons), and L_i is the unit social loss coefficient at demand site i (NTD/ton). The unit loss coefficients assigned to each demand sector are listed in Table 4.

The unit loss coefficients reflect the relative socioeconomic impact of water shortages across demand sectors, consistent with the supply priority structure adopted in regional water management practice in Taiwan. Municipal demand in Taipei City is assigned the highest unit loss (NT\$20/ton), reflecting its status as the national political and economic center with the highest population density and economic activity in the study area. Taoyuan City is assigned an intermediate value (NT\$15/ton), reflecting its substantial industrial base and municipal demand. The BanHsin service area and agricultural irrigation are each assigned NT\$10/ton, consistent with their lower per-unit economic sensitivity to water supply interruptions and with the established policy precedent of prioritizing municipal over agricultural demand during drought in Taiwan. These coefficients are treated as fixed parameters for the purpose of comparative scenario analysis; their absolute magnitudes do not affect the directional conclusions of the study. The robustness of the social welfare rankings to alternative loss structures is discussed in Sect. 4.

Financial cost

Financial cost is defined as the net economic burden incurred by each water utility under a given allocation scheme. It comprises three terms: revenue loss from unmet demand, expenditure on raw water purchased from the reservoir, and the transfer payment associated with BanHsin Diversion operations. Because the two

Table 4 Unit economic loss of each water demand sector

Demand Sector	Unit loss (NTD/ton)
Agricultural irrigation	10
Taipei City (M&I)	20
Taoyuan City (M&I)	15
BanHsin service area (M&I), New Taipei City	10

Table 5 Unit price parameters for TWC and TWD

Unit: NTD/ton	Taiwan Water Corporation (TWC)	Taipei Water Department (TWD)
$P1$	12	10
$P2$	2.09	0.3
$P3$ (≤ 30 tons)	5.3	-
$P3$ (> 30 tons)	5.9	-

For TWC, which supplies the BanHsin service area in New Taipei City, Taoyuan City, and the agricultural sector primarily from Shihmen Reservoir, the financial cost (FC) is expressed as:

utilities operate under different institutional and financial arrangements, the financial cost is formulated separately for each.

Three unit price parameters govern the financial cost calculation. $P1$ is the unit selling price of water to end users, representing the revenue foregone by the utility for each ton of demand that cannot be met. $P2$ is the unit cost of purchasing raw water from the reservoir, representing the direct expenditure incurred by the utility for each ton of water drawn from its primary supply source. $P3$ is the unit transfer price for inter-system water conveyance via the BanHsin Diversion Plant, representing the payment from the receiving utility (TWC) to the supplying utility (TWD) for each ton of water transferred between the two reservoir systems. The values of these parameters under current pricing are listed in Table 5.

$$FC_{(TWC)} = (\Delta q_{(Agri)} + \Delta q_{(Taoyuan)} + \Delta q_{(BanHsin)}) \times P1_{(TWC)} + Q_{(Shihmen)} \times P2_{(TWC)} + D \times P3 \quad (2)$$

where the first term is the revenue loss from unmet demand across all sectors served by TWC, calculated as the total volumetric shortfall multiplied by the unit selling price $P1$; the second term is the expenditure on raw water purchased from Shihmen Reservoir, calculated as the supply volume multiplied by the unit raw water cost $P2$; and the third term is the transfer payment made to TWD for water conveyed through the BanHsin Diversion Plant, where D is the volume of water transferred (million tons). The transfer payment $P3$ follows a two-tier pricing structure: the first 30 million tons per year are charged at NT\$5.3/ton, and any volume exceeding this threshold is charged at NT\$5.9/ton.

For TWD, which supplies Taipei City primarily from Feitsui Reservoir, the financial cost is expressed as:

$$FC_{(TWD)} = \Delta q_{(Taipei)} \times P1_{(TWD)} + Q_{(Feitsui)} \times P2_{(TWD)} - D \times P3 \quad (3)$$

where the first term is the revenue loss from unmet demand in Taipei City; the second term is the expenditure on raw water purchased from Feitsui Reservoir; and the third term is the transfer payment received from TWC for BanHsin Diversion operations. Because TWD is the supplier in this transaction, the transfer payment enters as revenue (negative cost) rather than expenditure. The raw water cost for TWD ($P2 = \text{NT\$}0.3/\text{ton}$) is substantially lower than that for TWC ($P2 = \text{NT\$}2.09/\text{ton}$), reflecting TWD's historical water rights arrangement within the Feitsui system.

Each utility is assumed to select the allocation scheme that minimizes its own financial cost. Because the two utilities operate under different cost structures and serve different demand areas, the scheme preferred by one utility may not coincide with the preference of the other, nor with the socially preferred outcome identified through the social loss criterion.

Under current pricing, TWC incurs a raw water cost of NT\$2.09/ton when drawing from Shihmen Reservoir, but a transfer price exceeding NT\$5/ton when obtaining water via the BanHsin Diversion Plant from the Feitsui system. This cost differential creates a financial disincentive for TWC to activate the BanHsin Diversion Plant, even when doing so would reduce system-wide social losses. Section 2.5 examines how adjustments to the unit selling price $P1$, the unit raw water cost $P2$, and the unit transfer price $P3$ can modify this financial calculus and align utility-level cost-minimizing behavior with the socially preferred allocation outcome across a range of hydrological scenarios.

Pricing sensitivity analysis

The water allocation outcomes simulated by WEAP under each scheme-scenario combination (Sect. 2.3) determine the physical quantities of water shortfalls, reservoir supply, and inter-system transfers. These physical outcomes are fixed for a given scheme and scenario and do not change when unit prices are adjusted. However, the financial cost incurred by each utility (Eq. 2) depends directly on the unit price parameters $P1$, $P2$, and $P3$. Changing these prices therefore alters the relative financial attractiveness of each allocation scheme without changing the underlying water distribution. It should be noted that this constitutes an institutional financial sensitivity exercise rather than a full demand-response water-pricing model: end-user consumption is held constant across all price levels, and the analysis does not capture behavioral adjustments in water demand that price changes might induce in practice.

This property enables a pricing sensitivity analysis. By systematically varying each unit-price parameter while holding the WEAP-simulated physical

allocations constant, the study examines how the financial cost ranking of the five allocation schemes responds to price changes from each utility's perspective. The objective is to identify price levels at which the scheme that minimizes each utility's financial cost coincides with the scheme that minimizes total social loss. In other words, the analysis seeks the conditions under which institutional self-interest aligns with the socially preferred outcome.

As defined in Sect. 2.4.2, there are five adjustable price parameters: the unit selling price $P1$ and the unit raw water cost $P2$ for each utility (four parameters), and the unit transfer price $P3$, which applies to both utilities simultaneously. The analysis proceeds in three steps. First, the relationship between each price parameter and the financial cost incurred by each utility is examined across all five scenarios, varying one parameter at a time. Second, the direction of price adjustment that brings each utility's preferred scheme closer to the socially preferred scheme is identified. Third, a combined pricing policy adjusting all parameters simultaneously is proposed and evaluated across all scenarios.

Because the analysis varies price parameters one at a time against fixed physical allocations, it is analogous to a one-at-a-time sensitivity analysis applied to the institutional decision framework rather than to the hydrological model. This approach reveals which price parameters most effectively influence utility behavior and under which hydrological conditions pricing-based alignment is achievable. It also identifies the boundary conditions under which no feasible price adjustment can reconcile institutional incentives with social welfare. In particular, when severe drought affects the Feitsui system, pricing-based alignment becomes unachievable, indicating the need for direct governmental intervention.

Results

This section presents the results of the scenario-based analysis in three parts. Section 3.1 summarizes the WEAP simulation outcomes for the 25 scheme-scenario combinations. Section 3.2 evaluates the social loss and financial cost associated with each scheme under each scenario, identifying where institutional preferences diverge from socially preferred outcomes. Section 3.3 presents the pricing sensitivity analysis, examining how adjustments to $P1$, $P2$, and $P3$ affect the alignment between utility-level and societal objectives.

WEAP simulation results

The WEAP model was run for all 25 scheme-scenario combinations defined in Sect. 2.3. The complete simulation results, including water deficiency, deficiency ratio, reservoir supply, and BanHsin Diversion transfer volumes, are provided in [Appendix Table 9](#). This section summarizes the key allocation patterns that emerge across scenarios.

Under Scenarios 1 through 3, which represent normal conditions and drought stress concentrated in the Shihmen system, Taipei City experienced no water shortages under any scheme, consistent with the historical reliability of the Feitsui system. In contrast, the Shihmen-served sectors (agriculture, Taoyuan City, and the BanHsin service area) experienced increasing deficiencies as drought severity increased. Schemes II and III, which assign supply priority to the Feitsui system for the BanHsin service area, substantially reduced these deficiencies by transferring supply burden from the Shihmen system to the Feitsui system through the BanHsin Diversion Plant. Under Scenario 3 (severe Shihmen drought) with Scheme III (Stage II diversion capacity, Feitsui priority), total water deficiency in the Shihmen-served sectors was reduced by over 80% compared to Scheme I (no diversion). Schemes IV and V, which retain Shihmen priority and activate Feitsui supply only as a backup, produced substantially smaller reductions in deficiency.

Under Scenarios 4 and 5, which introduce drought stress in the Feitsui system, the allocation dynamics change fundamentally. Water shortages now appear in Taipei City, which was previously unaffected. Schemes II and III continue to reduce shortages in the Shihmen-served sectors, but at the cost of increasing shortages in Taipei City. As drought severity in the Feitsui system increases from Scenario 4 to Scenario 5, this trade-off becomes more pronounced: Scheme III under Scenario 5 produces Taipei City shortages of approximately 152 million tons, compared to 135 million tons under Scheme I. The redistribution of supply through the BanHsin Diversion, which improved system-wide outcomes under Shihmen-dominated drought, now shifts scarcity toward the highest-priority demand center.

Table 6 Social loss and financial cost under current pricing (unit: million NTD)

	Scheme I	Scheme II	Scheme III	Scheme IV	Scheme V
Social Loss					
Scenario 1	929.40	456.85	50.80	807.75	782.70
Scenario 2	2731.07	1392.31	90.45	2286.44	2106.42
Scenario 3	3969.00	2509.24	599.09	3463.42	3351.38
Scenario 4	1185.36	859.37	674.87	1091.52	1073.32
Scenario 5	3780.03	3846.96	3888.40	3808.00	3808.73
TWC Financial Cost					
Scenario 1	3003.23	2908.92	2944.25	2951.99	2947.08
Scenario 2	4313.58	3726.38	3203.19	4128.33	4073.06
Scenario 3	5469.51	4801.70	3874.66	5249.76	5226.71
Scenario 4	3087.37	3071.11	3203.01	3038.64	3030.88
Scenario 5	3127.41	3110.56	3141.38	3094.19	3092.75
TWD Financial Cost					
Scenario 1	322.40	-108.40	-676.35	278.98	264.71
Scenario 2	322.40	-197.45	-1020.59	161.76	78.14
Scenario 3	322.40	-303.78	-1312.15	148.31	85.58
Scenario 4	399.12	-33.94	-524.80	369.50	362.69
Scenario 5	1633.24	1589.53	1507.81	1661.46	1662.58

Social loss and financial cost comparison

Table 6 presents the social loss and financial cost for each scheme under all five scenarios. The socially preferred scheme (minimum social loss) is Scheme III in Scenarios 1 through 4, where active redistribution of supply through the BanHsin Diversion Plant at full capacity substantially reduces aggregate shortages across the Shihmen-served sectors. Under Scenario 5 (severe Feitsui drought), however, Scheme I yields the lowest social loss. In this case, transferring water from the Feitsui system to the BanHsin service area intensifies shortages in Taipei City, where the unit social loss coefficient is the highest among all demand sectors (Table 4). The socially preferred outcome therefore shifts from maximizing inter-system transfer to protecting the most economically sensitive demand center.

From the perspective of TWC, the financially preferred scheme varies across scenarios. Under Scenarios 2 and 3, where drought reduces Shihmen supply, TWC's financial cost is minimized by Scheme III, because the reduction in revenue loss from unmet demand (via the unit selling price $P1$) outweighs the additional transfer payment (via the unit transfer price $P3$). In this case, TWC's self-interest coincides with the socially preferred outcome. Under Scenario 1, TWC prefers Scheme II, which utilizes the BanHsin Diversion at a lower capacity and thus incurs a smaller transfer payment. Under Scenarios 4 and 5, TWC prefers Scheme V, which retains Shihmen as the primary source and activates Feitsui supply only as a backup. In both cases, TWC's preference diverges from the socially preferred Scheme III.

From the perspective of TWD, Scheme III is financially preferred in nearly all scenarios, because the transfer payment received from TWC (via the unit transfer price $P3$) generates net revenue that exceeds the marginal raw water cost (via the unit raw water cost $P2$). However, under Scenario 5, the cost difference between Scheme III and other schemes narrows substantially, as increasing water shortages in Taipei City raise TWD's revenue loss (via the unit selling price $P1$), partially offsetting the transfer revenue.

Table 6 highlights the preferred scheme for each stakeholder under each scenario (shown in bold). The interests of both utilities align with the socially preferred outcome only under Scenarios 2 and 3, when drought is concentrated in the Shihmen system. Under normal conditions (Scenario 1) and under drought in the Feitsui system (Scenarios 4 and 5), at least one utility's preferred scheme diverges from the social optimum under current pricing. This persistent misalignment motivates the pricing sensitivity analysis presented in Sect. 3.3.

Pricing sensitivity analysis results

Figures 3, 4, 5, 6 and 7 present the results of the pricing sensitivity analysis described in Sect. 2.5. Each figure shows how the financial cost of a given utility responds to changes in a single price parameter across all five

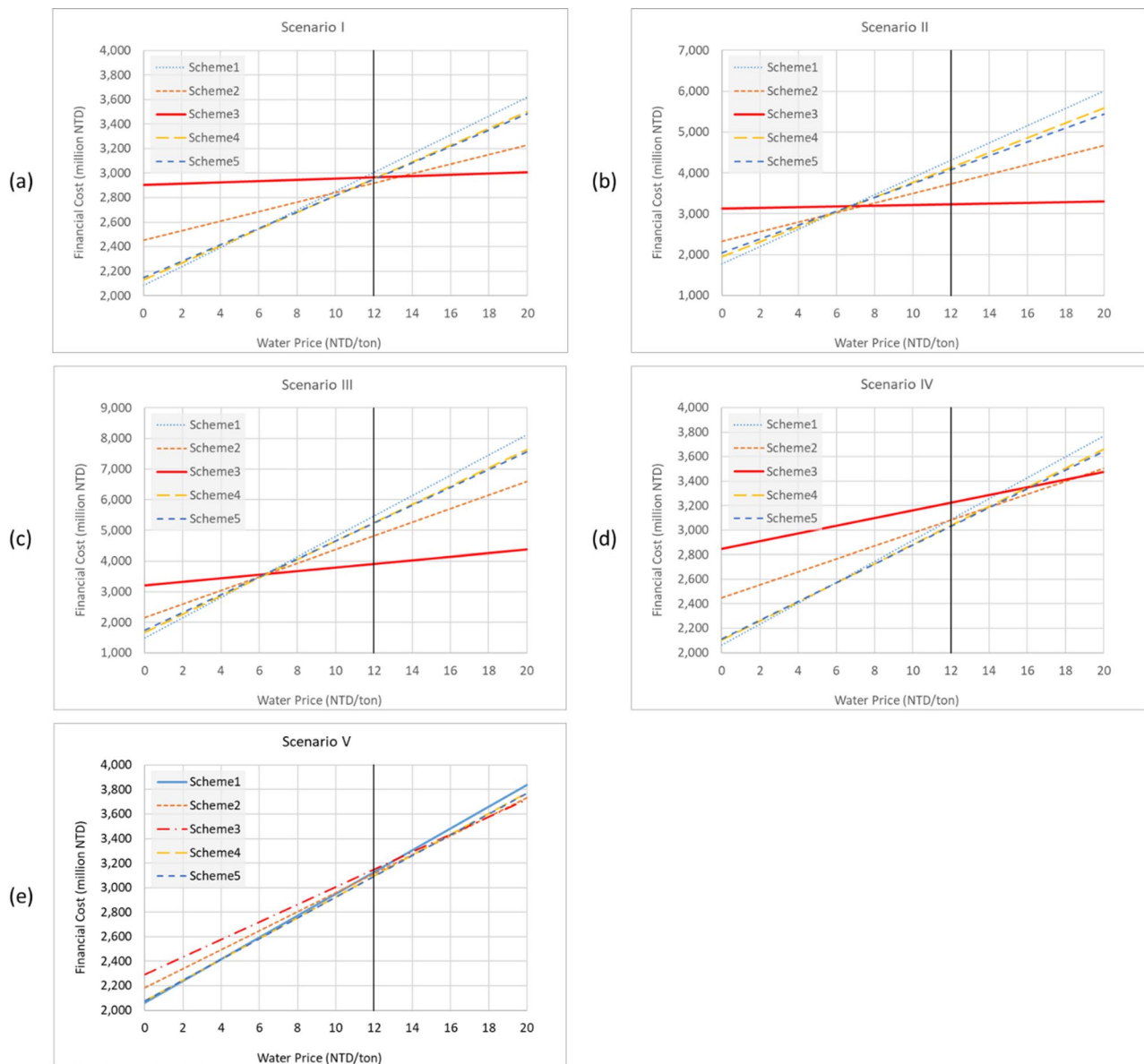


Fig. 3 Sensitivity of TWC's financial cost to changes in the unit selling price $P1$ across five hydrological scenarios (a-e). The bold red line indicates the socially preferred scheme. The vertical black line marks the current price level. Alignment is achieved when the bold line falls below all other lines

hydrological scenarios. Within each subplot, the bold red line represents the socially preferred scheme, and the vertical black line marks the current price level. Alignment between institutional and social preferences occurs when the bold line lies below all other lines at a given price level.

Unit selling price $P1$

Figure 3 presents the sensitivity of TWC's financial cost to changes in the unit selling price $P1$. Under the current price ($P1 = \text{NT\$}12/\text{ton}$), TWC's least-cost scheme conflicts with the socially preferred Scheme III in Scenarios 1, 4, and 5. As $P1$ increases, the revenue loss from unmet demand rises, increasing TWC's incentive to adopt schemes that minimize

water shortages. When $P1$ reaches approximately NT\\$20/ton, TWC's preferred scheme aligns with the socially preferred outcome in all scenarios except Scenario 5.

Figure 4 presents the corresponding analysis for TWD. Under Scenarios 1 through 3, no water shortage occurs in Taipei City, and all scheme lines overlap because changes in the unit selling price $P1$ do not affect TWD's financial cost. Under Scenario 4, TWD's preferred scheme already coincides with the social optimum at all price levels. Under Scenario 5, alignment is achieved when $P1$ exceeds approximately NT\\$18/ton, as the threat of revenue loss from Taipei City shortages discourages TWD from supporting large-scale inter-system transfers during severe Feitsui drought.

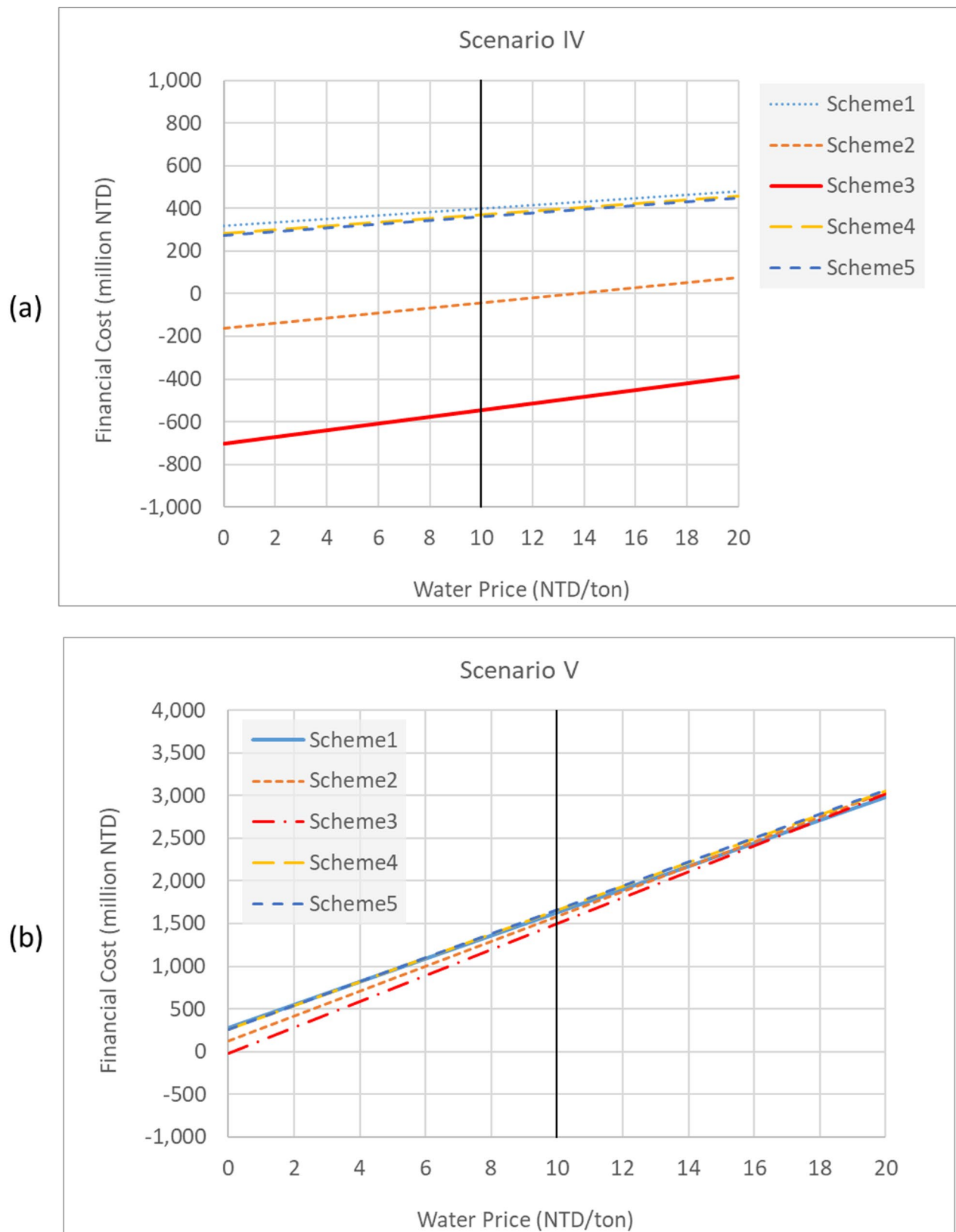


Fig. 4 Sensitivity of TWD's financial cost to changes in the unit selling price $P1$ across five hydrological scenarios (a–e). Format as in Fig. 3. Note: under Scenarios 1–3, no water shortage occurs in Taipei City, and all scheme lines overlap because adjustments to $P1$ do not affect TWD's financial cost

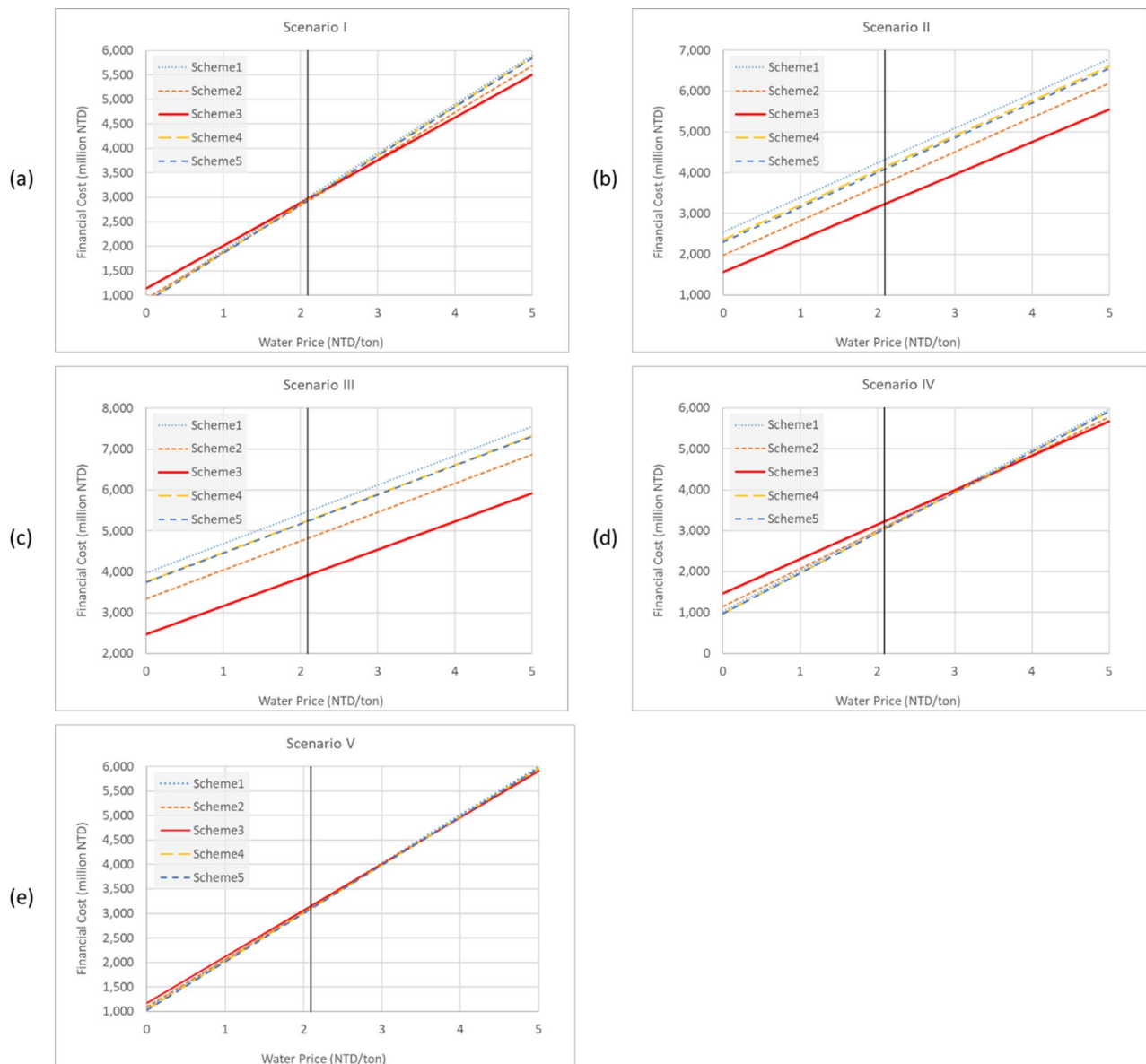


Fig. 5 Sensitivity of TWC's financial cost to changes in the unit raw water cost P_2 across five hydrological scenarios (a-e). Format as in Fig. 3

Unit raw water cost P_2

Figure 5 presents the sensitivity of TWC's financial cost to changes in the unit raw water cost P_2 . Increasing P_2 raises the cost of purchasing water from Shihmen Reservoir, making inter-system transfer via the BanHsin Diversion relatively more attractive. This adjustment reconciles TWC's preferences with the social optimum in Scenario 4 and achieves alignment in Scenarios 1 through 4 when P_2 reaches approximately NT\$4 to 5/ton. However, under Scenario 5, variations in P_2 have only a limited influence on TWC's financial cost across all schemes, and alignment cannot be achieved through this parameter alone.

Figure 6 presents the corresponding analysis for TWD. The unit raw water cost P_2 for TWD (NT\$0.3/ton) is already very low, and adjustments to P_2 produce minimal

changes in TWD's financial cost. Consequently, adjusting P_2 for TWD is not an effective instrument for achieving alignment, particularly under Scenario 5.

Unit transfer price P_3

Figure 7 presents the sensitivity of both utilities' financial costs to changes in the unit transfer price P_3 , shown for TWC (upper group) and TWD (lower group) within each subplot. Adjusting the unit transfer price P_3 simultaneously affects both utilities in opposite directions: reducing P_3 lowers TWC's cost of utilizing the BanHsin Diversion but also reduces TWD's revenue from the transfer.

Under Scenarios 1 through 3, reducing P_3 brings TWC's preferred scheme closer to the social optimum. Under

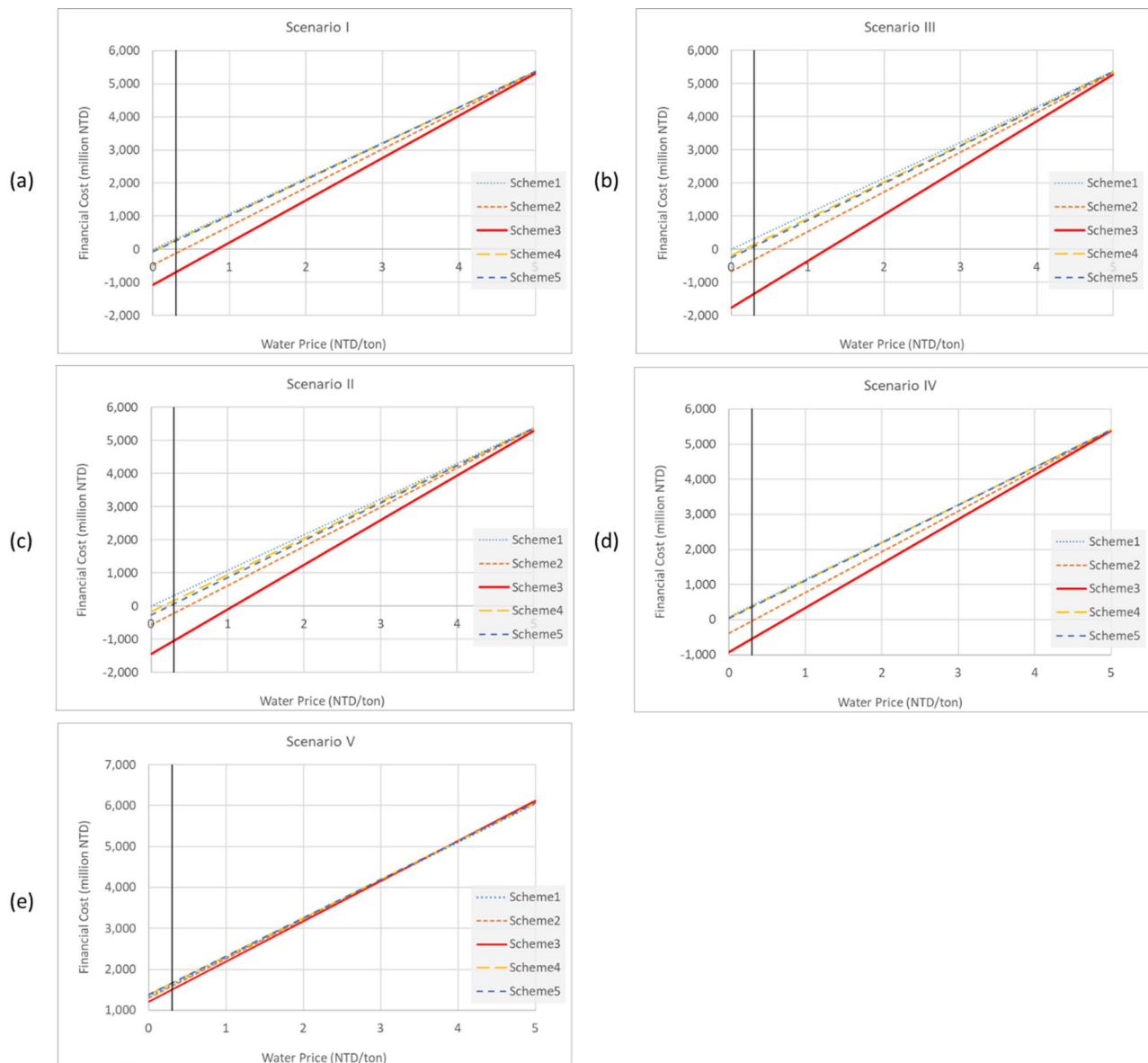


Fig. 6 Sensitivity of TWD's financial cost to changes in the unit raw water cost P_2 across five hydrological scenarios (a–e). The bold red line indicates the socially preferred scheme. The vertical black line marks the current price level. Alignment is achieved when the bold line falls below all other lines

Scenario 4, a reduction in P_3 achieves alignment for TWC. For TWD, reducing P_3 narrows the revenue advantage of Scheme III but does not change TWD's preferred scheme in Scenarios 1 through 4. Under Scenario 5, reducing P_3 achieves alignment for TWD (whose preferred scheme shifts from Scheme III toward Scheme I), but TWC remains misaligned.

Combined pricing policy

Based on the directional trends identified above, a combined pricing policy is proposed in which the unit selling price P_1 is increased, the unit raw water cost P_2 for TWC is increased, and the unit transfer price P_3 is reduced. The specific adjusted values are listed in Table 7. Under the

adjusted policy, the two-tier structure of P_3 is replaced by a single uniform rate of NT\$4.2/ton.

Table 8 presents the social loss and financial cost under the adjusted pricing policy. The preferred scheme for each stakeholder under each scenario is shown in bold. Under the adjusted pricing, both TWC and TWD prefer Scheme III in Scenarios 1 through 4, fully aligned with the socially preferred outcome. This represents a substantial improvement from the current pricing, under which alignment occurs only in Scenarios 2 and 3.

Under Scenario 5, TWD's preference shifts to Scheme I (consistent with the social optimum), but TWC continues to prefer Scheme III. This remaining misalignment reflects the structural limit of pricing-based coordination identified

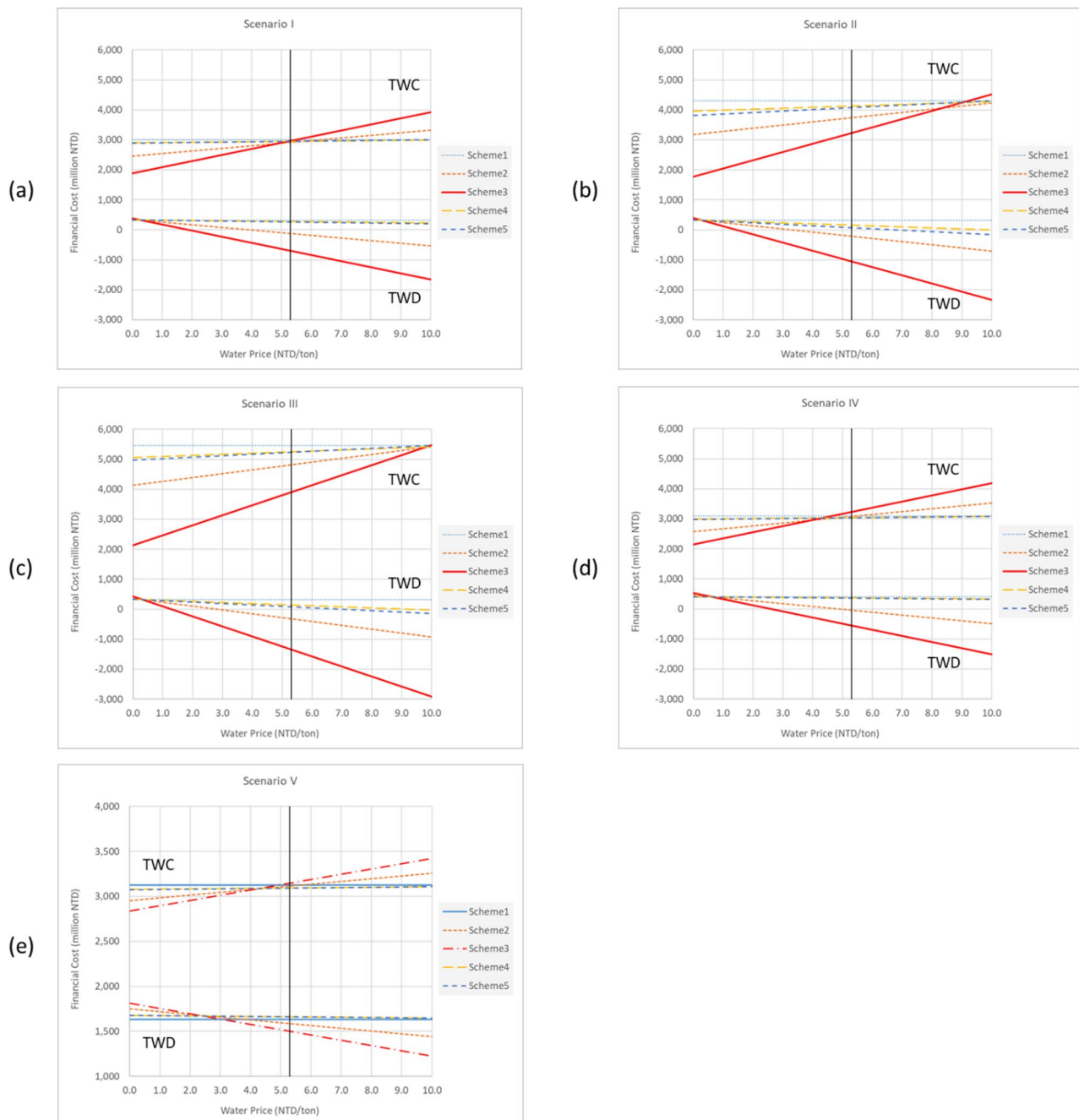


Fig. 7 Sensitivity of financial cost to changes in the unit transfer price P_3 across five hydrological scenarios (a–e), shown for TWC (upper group) and TWD (lower group) within each subplot. Format as in Fig. 3

Table 7 Adjusted unit price parameters (unit: NTD/ton)

Price Parameter	Utility	Before	After
Unit selling price P_1	TWC	12	20
Unit selling price P_1	TWD	10	20
Unit raw water cost P_2	TWC	2.09	3.8
Unit raw water cost P_2	TWD	0.3	0.3
Unit transfer price P_3	TWC/TWD	5.3 / 5.9	4.2

in Sect. 2.5: when severe drought affects the Feitsui system, no feasible combination of price adjustments can simultaneously satisfy all stakeholders, indicating that direct governmental intervention would be required under such conditions.

Table 8 Social loss and financial cost under adjusted pricing (unit: million NTD)

	Scheme I	Scheme II	Scheme III	Scheme IV	Scheme V
Social Loss					
Scenario 1	929.4	456.85	50.80	807.75	782.70
Scenario 2	2731.08	1392.31	90.45	2286.44	2106.42
Scenario 3	3969.00	2509.24	599.09	3463.42	3351.38
Scenario 4	1185.36	859.37	674.87	1091.52	1073.32
Scenario 5	3780.03	3846.96	3888.40	3808.00	3808.73
TWC Financial Cost					
Scenario 1	5321.03	4756.30	4273.00	5195.90	5174.70
Scenario 2	7458.38	6004.24	4367.23	7003.03	6834.69
Scenario 3	9341.93	7655.70	5186.17	8816.82	8721.21
Scenario 4	5458.45	4983.12	4694.75	5342.17	5322.39
Scenario 5	5524.70	5354.53	5273.74	5455.04	5452.36
TWD Financial Cost					
Scenario 1	322.40	-20.31	-472.35	288.01	276.65
Scenario 2	322.40	-91.19	-746.34	194.71	128.15
Scenario 3	322.40	-175.82	-978.40	184.00	134.07
Scenario 4	478.22	180.10	-163.70	465.52	460.97
Scenario 5	2984.63	3080.53	3084.59	3061.28	3063.92

Discussion

This study proposes a scenario-based policy analysis to examine how alternative water allocation schemes and pricing structures interact under varying drought stress patterns in a dual-reservoir urban water system. WEAP serves as a deterministic allocation framework that ensures internally consistent physical water distribution across scenarios, while institutional incentives and social welfare implications are interpreted externally based on the resulting shortages and costs. The discussion therefore focuses on why certain allocation and pricing arrangements perform differently across scenarios, rather than on the mechanics of the simulation itself.

A key insight emerging from the results is that institutional performance is governed less by the overall severity of drought than by where scarcity occurs within the system. When drought stress is concentrated in the Shihmen system, the Feitsui system continues to function as a relatively reliable source, enabling inter-system transfers that reduce aggregate social losses (Table 6). Under these conditions, allocation schemes that actively utilize the BanHsin Diversion Plant redistribute scarcity spatially rather than intensifying it locally. This explains why socially preferred outcomes consistently favor schemes with stronger interconnection under Scenarios 1 through 3, despite the absence of any increase in total water availability. The implication is that physical connectivity alone does not guarantee cooperative outcomes; rather, its effectiveness depends on the underlying asymmetry in supply reliability.

The pricing sensitivity analysis further reveals how this asymmetry shapes institutional incentives. Under Shihmen-dominated drought scenarios, adjustments to the unit

selling price $P1$ and the unit raw water cost $P2$ alter the relative financial burden faced by TWC and TWD, narrowing the gap between cost-minimizing behavior and socially preferred allocation schemes (Table 6). The cost-price relationships illustrated in Figs. 3 and 5 show that, within this regime, pricing can function as an indirect coordination mechanism by internalizing the consequences of localized shortages. Importantly, this coordination does not arise because pricing changes the physical feasibility of transfers, but because it reshapes the decision environment in which allocation choices are made. When combined into a single adjusted pricing policy (Table 7), the alignment between institutional preferences and social welfare is achieved in Scenarios 1 through 4, reducing the number of misaligned scenarios from three under current pricing to one (Table 8).

However, this mechanism exhibits a clear structural limit when drought stress shifts to the Feitsui system. As shown in Appendix Table 9, increased reliance on Feitsui supply under Scenarios 4 and 5 leads to substantial shortages in Taipei City, where the unit social loss coefficient is highest. In this regime, pricing adjustments are no longer able to fully reconcile institutional incentives with social welfare outcomes. Figures 4, 5, 6 and 7 indicate that, even with substantial price variation, the relative ranking of allocation schemes from the utilities' perspective remains partially misaligned with the socially preferred outcome under Scenario 5. This scenario therefore represents not an anomalous result, but a regime change in which scarcity affects the historically most reliable and highest-priority subsystem. Once this condition is reached, market-based coordination through pricing alone becomes insufficient.

This contrast between regimes highlights an important governance implication. Pricing mechanisms are effective when scarcity can be redistributed across subsystems without threatening core urban supply, but they lose traction when scarcity becomes systemic rather than localized. In such cases, reliance on decentralized, price-mediated decision-making risks amplifying social losses, even when institutions act rationally in pursuit of their financial objectives. The results thus suggest that pricing should be viewed as a conditional coordination tool, whose effectiveness depends on the spatial structure of scarcity and the relative vulnerability of demand centers.

The analytical framework employed in this study is transferable in structure to other multi-source urban water systems where decentralized utilities share interconnected infrastructure. The core elements of the approach, namely scenario-based allocation modeling, utility-specific financial cost formulation, and one-at-a-time pricing sensitivity analysis, do not depend on the specific institutional arrangements of the Taipei metropolitan system. However, the specific price parameters, unit loss coefficients, and threshold values at which alignment is achieved are context-dependent and would require adaptation to local governance

structures, cost accounting practices, and regulatory frameworks in other settings.

The interpretation of these findings should be considered within the scope of the study design. The scenarios examined represent stylized drought stress conditions rather than probabilistic projections, and the social loss estimates depend on priority-based coefficients that reflect relative societal impacts (Table 4).

To examine whether the social welfare rankings depend on the specific coefficient values assumed, three alternative structures were tested: reducing the Taipei City coefficient from NT\$20 to NT\$15/ton (Alt-1), elevating both urban weights to NT\$25 and NT\$20/ton for Taipei and Taoyuan respectively (Alt-2), and reducing the agricultural coefficient from NT\$10 to NT\$5/ton (Alt-3). Under all three alternatives, Scheme III remains the socially preferred outcome in Scenarios 1 through 4, and Scheme I in Scenario 5. This consistency reflects the fact that the physical shortfall differences between schemes are sufficiently large that no moderate reweighting of sectors can reverse the ranking. The dominance of Scheme III under Shihmen-dominated drought and Scheme I under severe Feitsui drought is therefore driven by the volume of water deficits rather than by the particular weights assigned to individual sectors. Detailed social loss values under all coefficient structures are provided in [Appendix Table 10](#).

Conclusions

This study demonstrates that in multi-reservoir urban water systems operated by independent utilities, physical infrastructure for inter-system water transfer does not automatically lead to coordinated regional allocation. Using the northern Taiwan water system as a case study, the analysis shows that institutional cost structures can prevent utilities from activating available transfer capacity, even when doing so would substantially reduce system-wide water shortages.

The WEAP simulation results demonstrate that inter-system water transfers via the BanHsin Diversion Plant can substantially reduce regional water shortages when drought stress is concentrated in the Shihmen system, with total deficiency in Shihmen-served sectors reduced by over 80% under the most favorable allocation scheme. However, under current pricing, the cost differential between local reservoir supply and inter-system transfer discourages TWC from activating the diversion, creating a persistent gap between institutional self-interest and socially preferred allocation.

The pricing sensitivity analysis indicates that adjustments to the unit selling price $P1$, the unit raw water cost $P2$, and the unit transfer price $P3$ can close this gap under a wide range of hydrological conditions. When water scarcity is localized and overall system reliability is maintained, a combined pricing policy aligns the preferences of both utilities with the socially preferred outcome, reducing the number

of misaligned scenarios from three to one. In this context, pricing serves not merely as a cost-recovery tool, but as a policy instrument that mediates transboundary allocation conflicts by reshaping the financial calculus of decentralized operational decisions.

At the same time, the results reveal a fundamental limitation of price-based coordination. When severe drought affects the Feitsui system, which is historically the most reliable and serves the highest-priority demand center, pricing incentives alone are insufficient to prevent socially undesirable outcomes. Under such conditions, rational institutional responses may still exacerbate overall welfare losses, indicating that market-based instruments cannot fully resolve allocation conflicts once scarcity becomes systemic. This finding highlights the importance of distinguishing between situations in which decentralized coordination remains viable and those in which direct governmental intervention may be required.

More broadly, these findings emphasize that effective water governance in interconnected urban water supply systems must be explicitly conditioned on system state rather than applied uniformly across situations. The pricing sensitivity framework employed in this study is transferable to other multi-source systems where decentralized utilities share interconnected infrastructure, although the specific parameter values and alignment thresholds are context-dependent. In practice, the implementation of pricing adjustments is also shaped by political considerations and affordability concerns, particularly for vulnerable populations. In Taiwan's regulated water supply context, increases to the unit selling price $P1$ may encounter resistance from regulators and consumers, and tariff increases can place a disproportionate burden on lower-income households. These distributional considerations suggest that pricing reforms aimed at incentive alignment should be accompanied by complementary affordability safeguards as part of any practical policy design process.

Future extensions of this framework could incorporate climate-informed hydrological projections to capture evolving drought characteristics and nonstationary risk, enabling the evaluation of pricing and allocation strategies across a broader range of plausible future conditions. Probabilistic approaches to scenario design would also enable formal uncertainty and sensitivity analysis of the social loss coefficients and financial cost parameters. Such developments would support more robust policy design by linking economic instruments with adaptive operational guidance, thereby enhancing the capacity of urban water systems to manage increasing hydroclimatic uncertainty (Fig. 8).

Appendix

See Tables 9 and 10; Fig. 8.

Table 9 WEAP simulation results for all scheme–scenario combinations: water deficiency, deficiency ratio, and reservoir supply volumes in the Taipei metropolitan water supply system

Scenario	Scheme	Water Deficiency (million ton)				Water Deficiency Ratio				Water Supply from (million ton)				
		Agriculture	Taipei	Taoyuan	BanHsin	Agriculture	Taipei	Taoyuan	BanHsin	Shihmen Reservoir	Feitsui Reservoir	BanHsin from Feitsui Reservoir	BanHsin from Shihmen Reservoir	
1	I	44.1	–	22.78	9.78	0.13	–	0.06	0.03	996.79	1074.67	–	316.61	
	II	25	–	10.4	3.39	0.07	–	0.03	0.01	950.38	1162.54	87.88	238.71	
	III	4.75	–	0.22	–	0.01	–	–	–	873.12	1278.45	203.78	131.02	
	IV	44.1	–	22.78	1.67	0.13	–	0.06	–	996.79	1083.49	882	316.61	
	V	44.1	–	22.78	–	0.13	–	0.06	–	996.79	1086.4	11.73	316.61	
2	I	88.19	–	81.63	41.64	0.26	–	0.2	0.12	849.73	1074.67	–	272.49	
	II	73.58	–	31.92	11.85	0.21	–	0.08	0.04	845.23	1180.72	106.05	203.67	
	III	9.04	–	–	–	0.03	–	–	–	798.78	1348.7	274.04	60.76	
	IV	88.19	–	81.63	12	0.26	–	0.2	0.04	849.73	1107.41	32.74	272.49	
	V	88.19	–	81.63	–	0.26	–	0.2	–	849.73	1124.47	49.81	272.49	
3	I	200.57	–	89.72	41.17	0.58	–	0.22	0.12	713.9	1074.67	–	257.12	
	II	163.13	–	44.89	13.64	0.47	–	0.11	0.04	706.84	1202.42	127.75	167.8	
	III	54.56	–	3.56	–	0.16	–	0.01	–	690.2	1408.21	333.54	1.26	
	IV	200.55	–	89.72	7.47	0.58	–	0.22	0.02	713.91	1110.15	35.49	257.12	
	V	200.55	–	89.72	–	0.58	–	0.22	–	713.91	1122.96	48.29	257.12	
4	I	50.23	7.91	24.25	10.74	0.15	0.01	0.06	0.03	987.91	1066.76	–	315.32	
	II	33.47	11.78	13.81	5.46	0.1	0.01	0.03	0.02	927.68	1158.89	96	227.88	
	III	21.82	15.7	7.14	2.38	0.06	0.01	0.02	0.01	845.13	1262.9	203.93	127.03	
	IV	50.23	8.8	24.25	3.3	0.15	0.01	0.06	0.01	987.91	1073.64	7.77	315.32	
	V	50.23	8.88	24.25	1.99	0.15	0.01	0.06	0.01	987.91	1075.1	9.31	315.32	
5	I	51.02	135.14	24.67	13.13	0.15	0.13	0.06	0.04	986.37	939.53	–	315	
	II	46.83	146.03	20.26	10.27	0.14	0.14	0.05	0.03	968.28	959.08	30.45	288.3	
	III	43.8	151.8	18.66	8.97	0.13	0.14	0.05	0.03	947.21	981.41	58.54	262.6	
	IV	51.02	139.65	24.65	9.01	0.15	0.13	0.06	0.03	986.41	938.15	3.13	315.02	
	V	51.02	139.79	24.65	8.86	0.15	0.13	0.06	0.03	986.41	938.06	3.19	315.02	

Water deficiency is reported in million tons; deficiency ratio is dimensionless (deficiency divided by annual demand). Water supply from each reservoir and BanHsin Diversion transfer volumes are reported in million tons

Table 10 Social loss (million NTD) under four unit loss coefficient structures

Coefficient Structure	Scenario	Scheme I	Scheme II	Scheme III	Scheme IV	Scheme V
Original (Agri = 10, Taipei = 20, Taoyuan = 15, BanHsin = 15)	1	929.4	456.9	50.8	807.8	782.7
	2	2730.9	1392.3	90.4	2286.3	2106.3
	3	3969.0	2509.2	599.0	3463.3	3351.3
	4	1185.3	859.4	675.0	1091.5	1073.5
	5	3780.0	3846.8	3888.4	3808.1	3808.6
Alt-1 (Taipei = 15)	1	929.4	456.9	50.8	807.8	782.7
	2	2730.9	1392.3	90.4	2286.3	2106.3
	3	3969.0	2509.2	599.0	3463.3	3351.3
	4	1145.8	800.5	596.5	1047.5	1029.1
	5	3104.3	3116.7	3129.4	3109.8	3109.7
Alt-2 (Taipei = 25, Taoyuan = 20)	1	1043.3	508.9	51.9	921.6	896.6
	2	3139.1	1552.0	90.4	2694.5	2514.5
	3	4417.6	2733.7	616.8	3912.0	3799.9
	4	1346.1	987.3	789.2	1256.8	1239.2
	5	4579.0	4678.3	4740.8	4629.6	4630.9
Alt-3 (Agri = 5)	1	708.9	331.9	27.1	587.2	562.2
	2	2290.0	1024.5	45.2	1845.4	1665.4
	3	2966.2	1693.6	326.2	2460.6	2348.6
	4	934.2	692.0	565.9	840.4	822.4
	5	3524.9	3612.7	3669.4	3553.0	3553.5

Bold values indicate the socially preferred scheme (minimum social loss) in each row

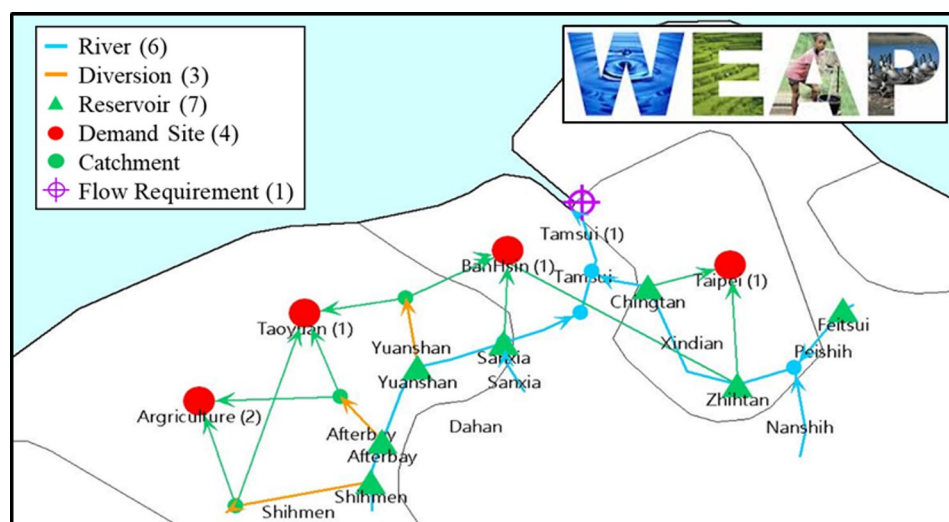


Fig. 8 Node-link configuration of the WEAP model for the northern Taiwan water supply system. The diagram shows six river reaches, seven reservoir and weir nodes, four demand sites, and one flow requirement node representing the minimum environmental flow at the Tamsui River outlet. Arrows indicate the direction of water conveyance, including the BanHsin Diversion linking the Feitsui and Shihmen systems

Acknowledgements

The authors thank the National Science and Technology Council, R.O.C., for support under grants MSTC-113-2625-M-002-017 and MSTC-113-2221-F-002-692.

Author contributions

C-Y.C. conceived the research idea, designed the methodology, and led the writing of the manuscript. Y-S.C. performed data collection and preprocessing, and contributed to the interpretation of results. Y-J.H. conducted the formal analysis and visualization, and assisted in drafting the results section. G J-Y.Y. supervised the entire project, provided critical revisions, and secured funding support. All authors reviewed and approved the final manuscript.

Funding

This work was supported by the National Science and Technology Council (NSTC), Taiwan. The authors were supported under grants MSTC-113-2625-M-002-017 and MSTC-113-2221-F-002-692.

Data availability

No datasets were generated or analysed during the current study.

Declarations

Competing interests

The authors declare no competing interests.

Received: 12 May 2025 / Accepted: 17 June 2026

Published online: 21 July 2026

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